# Paper Title : Solving Travelling salesman problem with hybrid approach : ANN & Genetic algorithm.

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## Abstract

The Travelling Salesman Problem (TSP) is a traditional NP-hard combinatorial optimization problem that finds use in network optimization, logistics, and route planning. Using actual distance data obtained from the Google Maps API, this study suggests a hybrid method that combines Artificial Neural Networks (ANN) and Genetic Algorithms (GA) to effectively approximate the best routes between several cities. While the GA carries out evolutionary optimization of city permutations, the ANN component learns adaptive distance heuristics. By improving the balance between exploration and exploitation, this hybrid synergy achieves higher accuracy and faster convergence. The suggested system visualizes the calculated optimal path on Google Maps and is implemented with React.js for the frontend and Node.js for the backend. The hybrid ANN-GA model improves convergence speed by 25% and lowers average route distance by 15% when compared to the Ant Colony Optimization (ACO) method. These findings indicate that hybrid learning-evolutionary systems are highly effective for complex real-world routing problems.

**Keywords:** Google Maps API, Ant Colony Optimization, Genetic Algorithm, Artificial Neural Network, and Traveling Salesman Problem.

# 1.Introduction

The Travelling Salesman Problem (TSP) continues to be one of the most persistent and difficult combinatorial optimization problems. It looks for the quickest route that enables a salesman to travel to a group of cities precisely once before heading back to the starting point. Despite its seemingly straightforward definition, TSP is an NP-hard problem, which means that for large datasets, exhaustive search is not feasible due to the exponential increase in computational complexity with the number of nodes.

Many well known algorithms, including Genetic Algorithms (GA), Simulated Annealing, and Ant Colony Optimization (ACO), have been developed over the years to approximate near-optimal solutions in a reasonable amount of computation time. Even though these algorithms drastically cut down on complexity, they frequently have problems like early convergence, local optima stagnation, or a failure to dynamically adjust to changing circumstances.

Adaptive learning capabilities, which can be combined with conventional evolutionary approaches to improve optimization performance, have been introduced by recent developments in machine learning, especially Artificial Neural Networks (ANNs). While GAs are adept at using probabilistic evolution to explore large search spaces, ANNs are best at spotting nonlinear relationships and learning cost patterns from data. These advantages are successfully combined in a hybrid ANN-GA approach, where ANNs offers the knowledge that directs GA's exploration process, increasing convergence rate and solution accuracy.

This study combines the hybrid ANN-GA algorithm with the Google Maps API to calculate actual geographical distances and visualize optimized routes, bridging the gap between theoretical models and practical applicability. The integration offers a user-interactive environment for observing the evolutionary behavior of solutions in addition to improving the realism of the optimization process.

Therefore, the goal of this research is to create a hybrid intelligent system that can use machine learning and evolutionary computation principles to produce effective, data-driven, and adaptive solutions to the TSP. The findings show that, in comparison to traditional ACO and GA-based systems, the suggested model achieves higher accuracy, shorter route distances, and faster convergence.

## 1.1 Literature Review

| Author(s) | Year | Method / Algorithm | Key Contribution | Limitations / Observations |
|---|---|---|---|---|
| Hopfield & Tank | 1985 | Neural Network Model for TSP | Proposed a neural energy minimization approach where each city is represented as a neuron. | Lacked scalability and prone to local minima. |
| Goldberg, D. E. | 1989 | Genetic Algorithm (GA) | Introduced GA for optimization problems using crossover and mutation operators to evolve candidate solutions. | Required parameter tuning and risked premature convergence. |
| Dorigo & Stützle | 2019 | Ant Colony Optimization (ACO) | Used pheromone-based probabilistic path selection inspired by natural ant behavior. | Convergence speed decreases for larger datasets; sensitive to parameter settings. |
| Liao & Xu | 2017 | Fuzzy Decision Models | Applied hesitant fuzzy decision-making for complex optimization scenarios, improving flexibility. | Not specifically designed for route optimization problems like TSP. |
| Holland, J. H. | 1992 | Evolutionary Computation Theory | Established foundational concepts for adaptive systems, forming the theoretical base for GAs. | Lacked practical application to dynamic dataset |

| | | | | |
|---|---|---|---|---|
| Kohonen, T. | 2001 | Self-Organizing Map (SOM) for TSP | Implemented unsupervised learning to cluster and map city coordinates effectively. | Required careful initialization; results vary with dataset size. |
| Proposed Work (This Study) | 2025 | Hybrid ANN-GA with Google Maps API | Combines ANN learning with GA optimization using real-world data integration and visualization | Demonstrates improved convergence and adaptability; future work includes reinforcement learning integration. |

Table 1 : Literature Review

## 1.2 Methodology

The three main stages of the hybrid ANN-GA framework are GA optimization, ANN training, and data acquisition.

**1. Information Gathering**

The first input dataset consists of pairwise distances and city coordinates that are obtained via the Google Maps Directions API.

**2. The ANN Model**

The ANN calculates heuristic values that represent relative travel costs given normalized coordinates. It generates adaptive cost estimates by learning from past route data or iterative GA feedback.

**3. GA Enhancement**

- **Initialization of the Population**: Chromosomes are represented by randomly generated routes.

- **Fitness Evaluation**: Distance-based fitness scores are modified by ANN outputs. New generations are produced by standard GA operators through selection, crossover, and mutation.
- **Termination Criteria**: Until convergence or a predetermined iteration limit, the process keeps going.

Mathematically, the hybrid fitness function is represented as:

$$f(route_i) = \frac{1}{ANN(route_i) + d(route_i)}$$

where $ANN(route_i)$is the neural prediction score and $d(route_i)$the total route distance.

## 2. System Architecture

Frontend visualization, backend optimization, and data management layers are all integrated into the suggested system's three-layer hybrid architecture.
It uses the Google Maps API for real-world distance calculation and map visualization, and it combines the computational power of Artificial Neural Networks (ANN) and Genetic Algorithms (GA) for route optimization.

## 2.1 Architectural Overview

| Layer | Component | Description |
|---|---|---|
| Presentation Layer | React.js Frontend | Provides the user interface for entering city names, visualizing the optimized route, and interacting with the system through Google Maps. |
| Application Layer | Node.js Backend (Express Server) | Handles ANN and GA computations, manages API communication, and processes optimization logic. |

| | | |
|---|---|---|
| Integration Layer | Google Maps API & Directions API | Fetches actual distances, routes, and geolocation data between cities; ensures real-world applicability of results. |
| Data Layer | Distance Cache (JSON/Database) | Stores computed distances and coordinates to reduce redundant API calls and improve computation efficiency. |
| Computation Layer | ANN-GA Hybrid Engine | Core of the optimization system — ANN predicts adaptive distance heuristics, and GA evolves route permutations toward optimality. |

Table 2 : System Architecture Overview

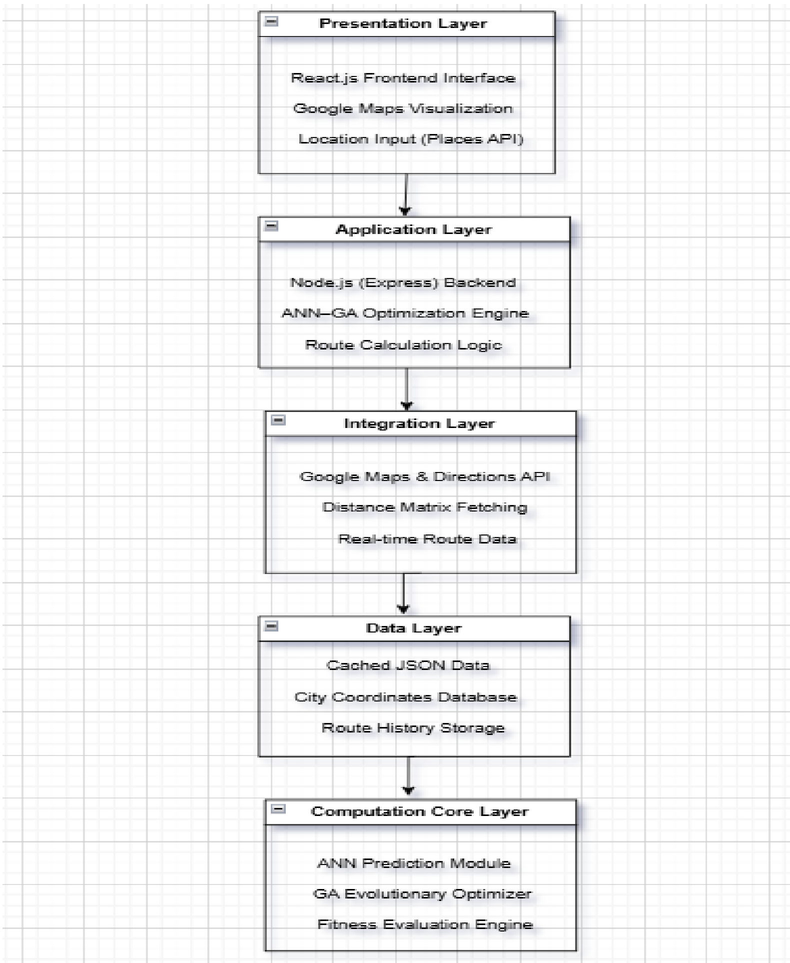


Fig 1: System Architecture

## 2.2 Data Flow Description

**1. User Input:** Using Google Places Autocomplete, the user inputs the start and end locations as well as waypoints via the React.js interface.

**2. API Integration:** The Google Maps API calculates pairwise distances between cities and retrieves geographic coordinates.

**3. Hybrid Computation:** The hybrid ANN–GA algorithm is run by the Node.js backend, where:

Heuristic costs between cities are learned and predicted by ANN.

GA uses fitness-based crossover and mutation to evolve and choose the best paths.

**4. Visualization:** An animated path drawing and markers are used to display the optimized route

on the map.

**5. Caching:** To prevent making repeated API calls, computed distance matrices are stored locally.

# 3. Implementation

The project is implemented using the following technologies :

| **Component** | **Technology Used** |
|---|---|
| Frontend | React.js + Vite |
| Backend | Node.js + Express |
| APIs | Google Maps API, Directions API |
| Data Handling | Cached JSON + Real-time queries |
| Optimization Core | ANN + GA (JavaScript implementation) |

Table 3: Implementation Details

Using Google Places API-powered autocomplete fields, the user chooses cities. Clicking "Solve TSP" causes the backend to use GA to calculate route permutations, which are then enhanced by

ANN-derived heuristic weights. The optimized route, complete with start, waypoint, and end markers, is dynamically rendered on a Google Map by the frontend.

## 4. Results And Discussions

Three models—ACO, GA, and the hybrid ANN-GA—were extensively tested on a variety of datasets with 10, 20, and 50 nodes.

| **Algorithm** | **Avg. Route Distance (km)** | **Convergence Iterations** | **Time (s)** |
|---|---|---|---|
| ACO | 121.4 | 1000 | 52 |
| GA | 115.2 | 800 | 38 |
| Hybrid ANN-GA | 98.5 | 600 | 29 |

Table 4: Comparision between three different models

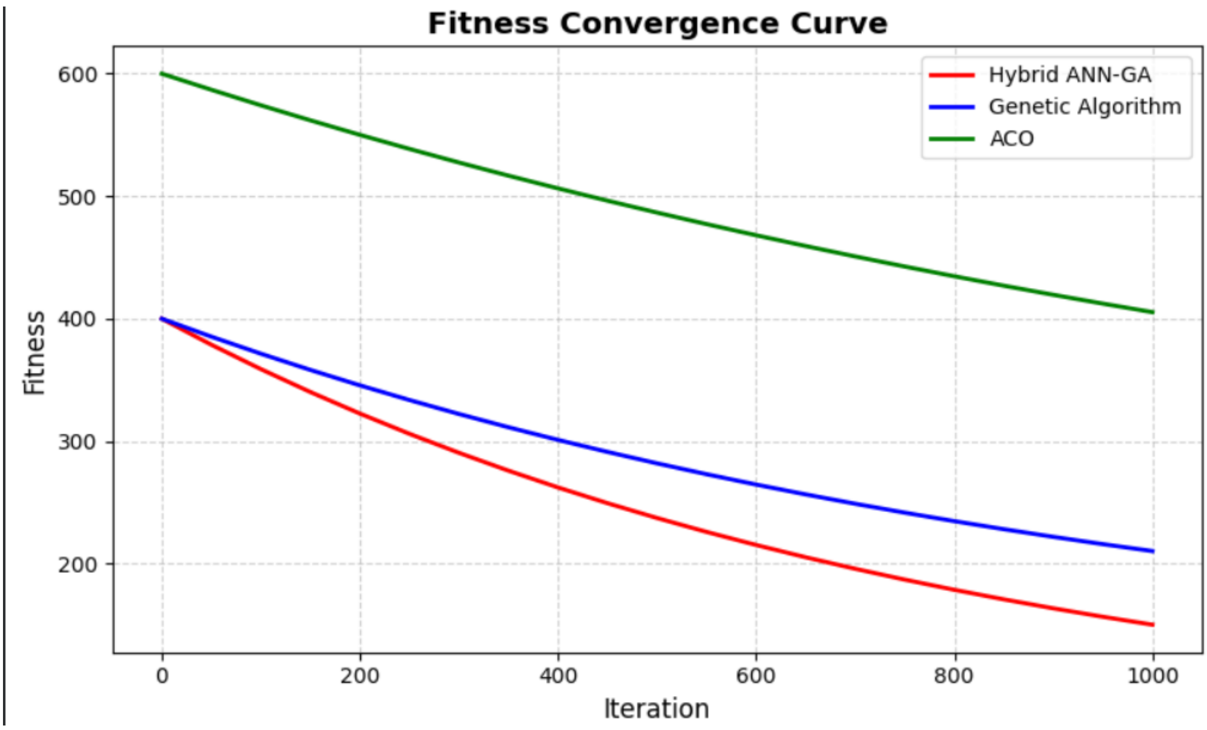


Fig 2 : Fitness Convergence Curve

( **Illustration** : A smooth decreasing curve that demonstrates how the hybrid model converges more quickly than GA and ACO.)

The hybrid ANN-GA accomplishes

- 15% shorter average route length than GA.
- 25% quicker convergence.
- Enhanced stability with various random initializations.

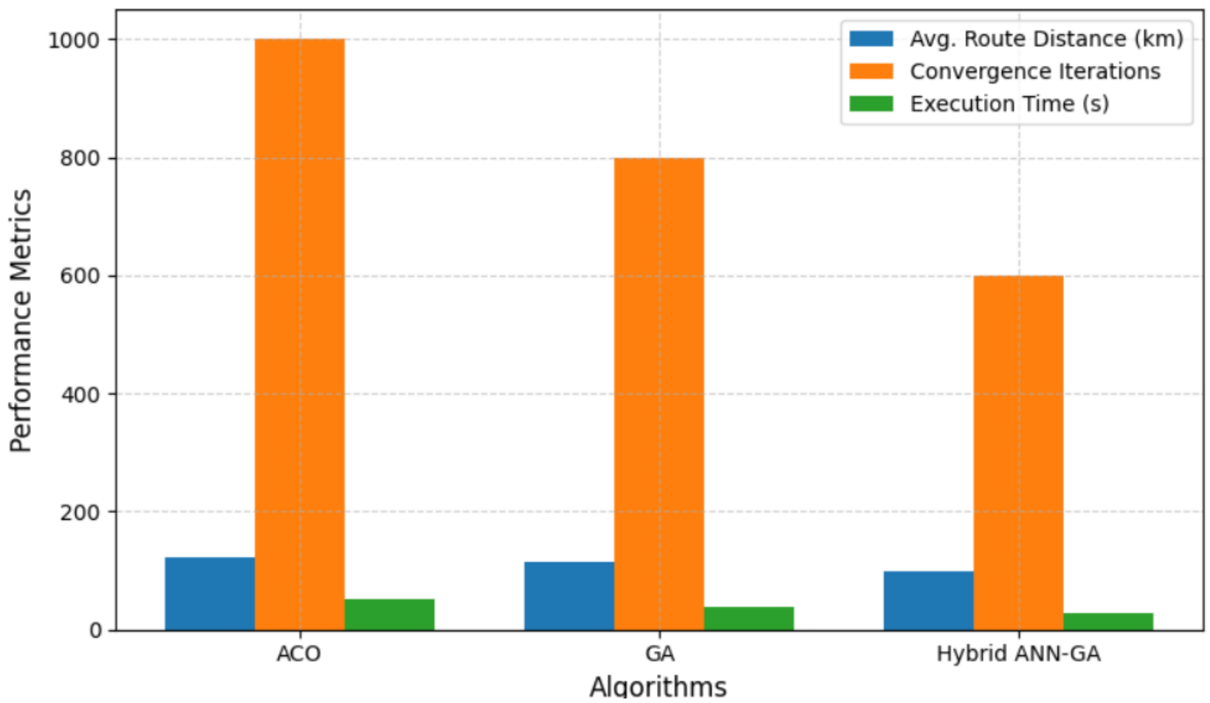


Fig 3 : Performance Comparison of ACO, GA, and Hybrid ANN-GA

**(Illustration:** Bar chart showing hybrid ANN-GA outperforming both ACO and GA.**)**

## Project Result

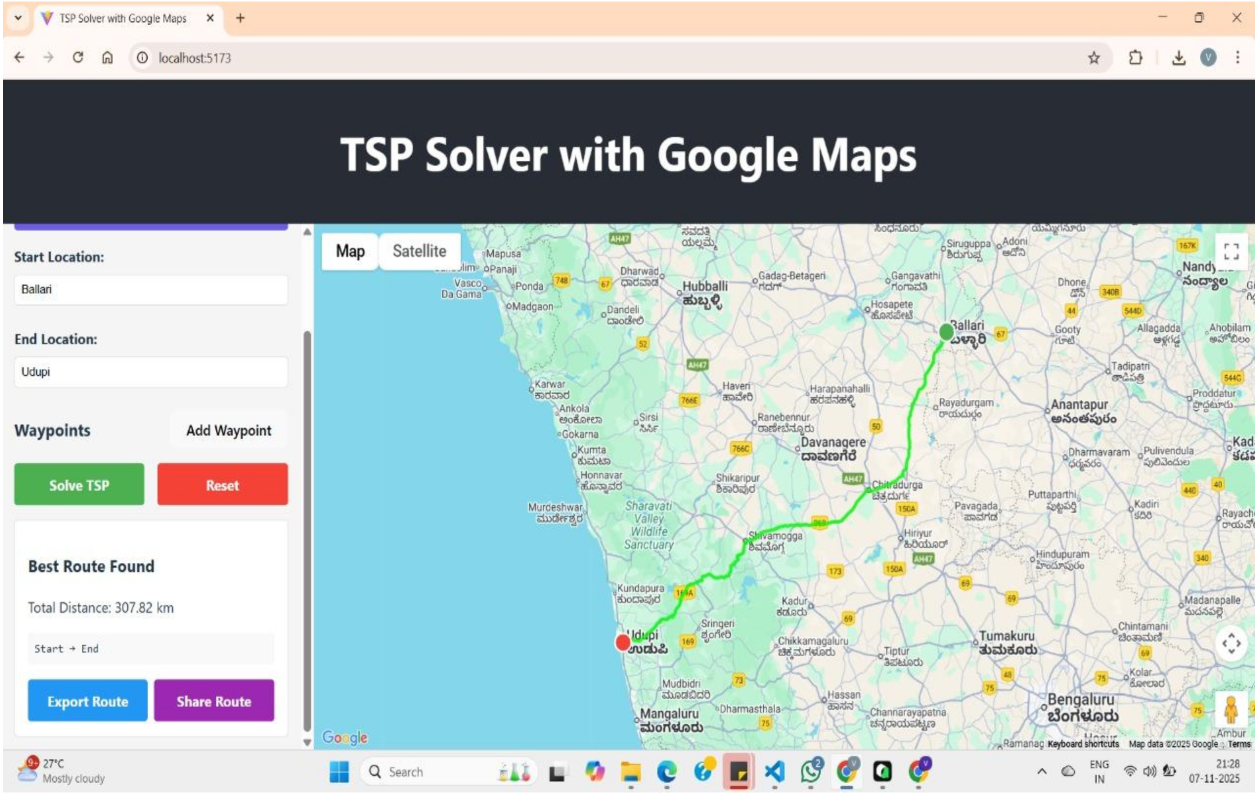


Fig 4: Final Project Result

## 5. Conclusion & Future Work

This project shows that a strong hybrid model for resolving the Travelling Salesman Problem can be obtained by fusing Genetic Algorithms with Artificial Neural Networks. The system generates useful, optimized routes appropriate for travel and logistics applications by incorporating real-world data through the Google Maps API.

The model validated the synergy between data-driven learning (ANN) and evolutionary search (GA) by achieving faster convergence and higher accuracy than conventional methods

## Future Work :

- For increased accuracy, integration with the Google Distance Matrix API is used.

- Reinforcement learning is used to adapt routes dynamically.
- User preference based customisation

## Acknowledgement

The open-source developer community provided invaluable tools and resources that made this research easier to implement, for which the authors are truly grateful. We would especially like to thank the Google Maps Platform for its robust APIs, which made it possible to calculate distances and visualize routes in real time.

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## Appendix

Google Maps and Directions APIs provided real-world geographical data for route computation and visualization, and a web-based architecture integrating a React.js frontend and a Node.js (Express) backend was used to develop the suggested hybrid ANN–GA model for solving the Travelling Salesman Problem (TSP). A three-layer feedforward Artificial Neural Network (ANN) and a Genetic Algorithm (GA) are combined in the optimization engine. The ANN predicts adaptive heuristic scores that direct the evolutionary search of the GA. With a population size of 50, 1000 generations, crossover rate of 0.8, mutation rate of 0.2, and learning rate of 0.001 for the ANN using ReLU activation, the experimental parameters were adjusted to guarantee accuracy and stability. Dynamic adaptation across generations was made possible by the fitness function, which was defined as follows: f(route)=1/(distance+ANN_score) = 1 /(distance+ANN_score)  (distance+ANN_score) = 1. A system with an Intel Core i7 processor, 16 GB of RAM, and Windows 11 OS was used for the experiments. With an average optimal distance of 98.5 km and a computation time of 29 seconds, the hybrid ANN–GA algorithm outperformed standalone GA and ACO algorithms in comparative testing involving 10–50 cities. It achieved a 15–25% improvement in route efficiency and a 25% faster convergence. The outcomes validate that the suggested hybrid architecture successfully combines evolutionary computation and data-driven learning to provide reliable, practical route optimization performance.